

2.7 - Using Formulas

A formula is an equation with more than one letter. Today you solve it for a DIFFERENT letter than the one it is written for.

Speak Like A Mathematician:

literal equation:

formula:

dimensional analysis:

THE 2.7 MOVE — every other letter is just a number you don't know yet

1. CIRCLE the letter you are solving for. That is the only thing you are trying to get alone.
2. Every other letter is a NUMBER. It does not have a value yet — treat it exactly like the 5 or the 3 in Chapter 2.
3. UNDO in reverse order: add / subtract first, then multiply / divide. (Same as 2.3.)
4. If your letter shows up in TWO terms, get both terms on one side and FACTOR it out. (Distributive Property, run backward.)
5. Done when your letter is alone on one side and does NOT appear on the other side.

Part 1 — Same moves, letters instead of numbers

The left column is 2.3. The right column is 2.7. Every step on the right is the SAME step as the one on the left.

Step	With numbers	With letters (solve for x)
Start	$3x + 5 = 20$	
Undo the + : subtract	$3x = 15$	
Undo the × : divide	$x = 5$	
What did you divide by?	3	

Part 2 — Example 1: Solve $P = 2l + 2w$ for w .

Worked for you. Read each line and say which step of the MOVE it is.

$$P = 2l + 2w \quad \text{circle } w; \text{ } 2l \text{ and } 2 \text{ are "numbers"}$$

$$P - 2l = 2w \quad \text{subtract } 2l \text{ from both sides}$$

$$(P - 2l) / 2 = w \quad \rightarrow \quad w = (P - 2l) / 2 \quad \text{divide both sides by } 2$$

Note: Keep the whole top together: $(P - 2l) / 2$. Dividing only the P by 2 is the #1 error in this section.

Part 3 — Example 2: Solve $A = \frac{1}{2}bh$ for h .

Half worked. Fill the blanks.

$$A = \frac{1}{2}bh \quad \text{multiply both sides by } 2: \underline{\hspace{2cm}}$$

$$\text{divide both sides by } b: \underline{\hspace{2cm}}$$

Part 4 — Example 3: Solve $y = mx + b$ for x .

Nothing filled in. Two undo steps — say them before you write them.

Part 5 — Example 4: the letter shows up twice. Solve $3x + ax = c$ for x .

You cannot get x alone by undoing — it is in two terms. FACTOR it out first, then divide by what is left in the parentheses.

$$3x + ax = c \quad \text{factor } x \text{ out of the left side: } \underline{\hspace{2cm}}$$

$$\text{divide both sides by } (3 + a): \underline{\hspace{2cm}}$$

Your turn: Solve $5y - ky = 8$ for y .

Part 6 — Example 5: Use the formula twice. $d = rt$ a) Solve $d = rt$ for t .

b) A bus travels 375 miles at 60 miles per hour. Use your formula from (a) to find the time.

c) Fahrenheit to Celsius. $F = (9/5)C + 32$. Solve for C , then convert 68°F .**Part 7 — Dimensional analysis: multiply by 1, change the units****UNIT RATIOS**

A unit ratio is a fraction equal to 1, like $60 \text{ min} / 1 \text{ h}$ or $1 \text{ h} / 60 \text{ min}$. Both are 1, so multiplying by either one does not change the VALUE.

Pick the one that puts the unit you want to CANCEL on the bottom. Cross the unit out like a common factor. What is left is your new unit.

Example 6 (worked): Convert 3.5 hours to minutes.

$$3.5 \text{ h} \times (60 \text{ min} / 1 \text{ h}) = 210 \text{ min} \quad \text{h on top, h on the bottom — cancel}$$

Example 7 (half worked): Convert 90 miles per hour to feet per second. ($1 \text{ mi} = 5280 \text{ ft}$, $1 \text{ h} = 3600 \text{ s}$)

$$90 \text{ mi/h} \times (5280 \text{ ft} / 1 \text{ mi}) \times (\rule{1.5cm}{0.4pt})$$

$$= (90 \times 5280) / 3600 = \rule{1.5cm}{0.4pt}$$

Example 8 (you): A sprinter runs 100 meters in 10 seconds. Find his speed in kilometers per hour. ($1 \text{ km} = 1000 \text{ m}$, $1 \text{ h} = 3600 \text{ s}$)

Exit — On your owna) Solve $V = lwh$ for h . minutes.b) Solve $S = 2\pi r^2 + 2\pi rh$ for h .

c) Convert 2 days to